Partial Differential Equations

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Introduction to Partial Differential Equations (PDE)

A **Partial Differential Equation (PDE)** is an equation involving partial derivatives of an unknown function of two or more independent variables.

If a function depends on several variables, its derivatives with respect to those variables are called **partial derivatives**.

For example, if

$$z = f(x, y)$$

then

$$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$$

are partial derivatives.

Definition of PDE

A Partial Differential Equation is an equation containing:

- dependent variables,
- independent variables,
- and partial derivatives of the dependent variable.

General Form

$$F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \dots\right) = 0$$

Examples of PDEs

First-order PDE

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$$

Second-order PDE

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

This equation is known as the Laplace equation.

Formation of PDE

A PDE is generally formed by:

- Taking partial derivatives of a given relation containing arbitrary constants or arbitrary functions.
- Eliminating those arbitrary constants/functions.

Example:

Given

$$z = ax + by$$

Differentiate:

$$\frac{\partial z}{\partial x} = a, \qquad \frac{\partial z}{\partial y} = b$$

Eliminate $a$and b:

$$z = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$$

Hence the PDE is

$$xp + yq = z$$

Where,

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}$$

Order of a PDE

The **order** of a PDE is the order of the highest partial derivative present.

Example

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial y} = 0$$

Highest derivative is second order.

Hence, the PDE is a **second-order PDE**.

Degree of a PDE

The **degree** of a PDE is the power of the highest order derivative after removing radicals and fractions involving derivatives.

Example

$$\left(\frac{\partial^2 z}{\partial x^2}\right)^3 + \frac{\partial z}{\partial y} = 0$$

Order = 2

Degree = 3

Classification of PDEs

1. Linear PDE

Dependent variable and its derivatives appear linearly.

Example:

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$$

2. Nonlinear PDE

Contains powers/products of derivatives.

Example:

$$\left(\frac{\partial z}{\partial x}\right)^2 + \frac{\partial z}{\partial y} = 0$$

Applications of PDE

Partial Differential Equations are widely used in:

- Heat transfer
- Fluid mechanics
- Electromagnetic theory
- Wave motion
- Quantum mechanics
- Image processing
- Engineering and physics

Common PDEs

- **Heat Equation**

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$$

- **Wave Equation**

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

- **Laplace Equation**

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Conclusion

Partial Differential Equations describe relationships involving functions of several variables and their partial derivatives. They play a fundamental role in mathematics, physics, and engineering for modeling real-world phenomena such as heat flow, vibrations, fluid dynamics, and electromagnetic fields.

1. Form the P.D.E by eliminating arbitrary constants a, b give $z = xy\sqrt{x^2 - a^2} + b$.

Solution: Given

$$z = xy\sqrt{x^2 - a^2} + b$$

where a and b are arbitrary constants.

We need to form the partial differential equation by eliminating a and b.

Let

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}$$

First differentiate partially with respect to y:

$$q = \frac{\partial}{\partial y}\left(xy\sqrt{x^2 - a^2} + b\right)$$

$$q = x\sqrt{x^2 - a^2}$$

Now square both sides:

$$q^2 = x^2(x^2 - a^2)$$

$$q^2 = x^4 - a^2x^2$$

Hence,

$$a^2 = x^2 - \frac{q^2}{x^2}$$

Now differentiate partially with respect to x:

$$p = \frac{\partial}{\partial x}\left(xy\sqrt{x^2 - a^2} + b\right)$$

Using product rule,

$$p = y\sqrt{x^2 - a^2} + xy\left(\frac{x}{\sqrt{x^2 - a^2}}\right)$$

$$p = y\left(\frac{x^2 - a^2 + x^2}{\sqrt{x^2 - a^2}}\right)$$

$$p = y\left(\frac{2x^2 - a^2}{\sqrt{x^2 - a^2}}\right)$$

From

$$q = x\sqrt{x^2 - a^2}$$

we get

$$\sqrt{x^2 - a^2} = \frac{q}{x}$$

Also,

$$a^2 = x^2 - \frac{q^2}{x^2}$$

Substitute in p:

$$p = y\frac{2x^2 - \left(x^2 - \frac{q^2}{x^2}\right)}{q/x}$$

$$p = y\frac{x^2 + \frac{q^2}{x^2}}{q/x}$$

$$p = \frac{y(x^4 + q^2)}{xq}$$

Therefore,

$$pxq = y(x^4 + q^2)$$

Hence the required P.D.E is

$$\boxed{x\,p\,q = y(x^4 + q^2)}$$

2. Construct the PDE corresponding to $z = (x - a)^2 + (y - b)^2$.

Solution: Given

$$z = (x - a)^2 + (y - b)^2$$

where a and b are arbitrary constants.

We must eliminate a and b to form the PDE.

Let

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

Differentiate partially with respect to x:

$$p = \frac{\partial}{\partial x}[(x - a)^2 + (y - b)^2]$$

$$p = 2(x - a)$$

Hence,

$$x - a = \frac{p}{2}$$

Now differentiate partially with respect to y:

$$q = \frac{\partial}{\partial y}[(x-a)^2 + (y-b)^2]$$

$$q = 2(y-b)$$

Hence,

$$y - b = \frac{q}{2}$$

Substitute these into the original equation:

$$z = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2$$

$$z = \frac{p^2 + q^2}{4}$$

Therefore,

$$\boxed{p^2 + q^2 = 4z}$$

Where,

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

This is the required PDE.

3. Determine the PDE corresponding to $z = ax^2 + by^2$.

Solution: Given

$$z = ax^2 + by^2$$

where a and b are arbitrary constants.

We need to eliminate a and b to form the PDE.

Let

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

Differentiate partially with respect to x:

$$p = \frac{\partial}{\partial x}(ax^2 + by^2)$$
$$p = 2ax$$

Hence,

$$a = \frac{p}{2x}$$

Now differentiate partially with respect to y:

$$q = \frac{\partial}{\partial y}(ax^2 + by^2)$$
$$q = 2by$$

Hence,

$$b = \frac{q}{2y}$$

Substitute these values into the original equation:

$$z = x^2\left(\frac{p}{2x}\right) + y^2\left(\frac{q}{2y}\right)$$
$$z = \frac{xp}{2} + \frac{yq}{2}$$

Therefore,

$$\boxed{xp + yq = 2z}$$

Where,

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

This is the required PDE.

4. Construct the P.D.E by eliminating arbitrary function $z = y^2 + 2f(\frac{1}{x} + logy)$.

Solution: Given

$$z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$$

Let

$$u = \frac{1}{x} + \log y$$

so that

$$z = y^2 + 2f(u)$$

We need to eliminate the arbitrary function f.

Let

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

Differentiate partially with respect to x:

$$p = 2f'(u)\frac{\partial u}{\partial x}$$

Since

$$\frac{\partial u}{\partial x} = -\frac{1}{x^2}$$

we get

$$p = -\frac{2f'(u)}{x^2}$$

Now differentiate partially with respect to y:

$$q = 2y + 2f'(u)\frac{\partial u}{\partial y}$$

and

$$\frac{\partial u}{\partial y} = \frac{1}{y}$$

Hence,

$$q = 2y + \frac{2f'(u)}{y}$$

From the equation for p,

$$2f'(u) = -px^2$$

Substitute into the equation for q:

$$q = 2y - \frac{px^2}{y}$$

Multiply by y:

$$yq = 2y^2 - px^2$$

Therefore,

$$\boxed{x^2p + yq = 2y^2}$$

where,

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

This is the required PDE.

5. Determine the PDE of $z = yf(x) + x\phi(y)$, where f & ϕ are arbitrary functions.

Solution: Given

$$z = yf(x) + x\phi(y)$$

where f and ϕ are arbitrary functions.

We must eliminate the arbitrary functions to form the PDE.

Let

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$$

Differentiate partially with respect to x:

$$p = yf'(x) + \phi(y)$$

Differentiate partially with respect to y:

$$q = f(x) + x\phi'(y)$$

Now differentiate again:

$$\frac{\partial p}{\partial y} = f'(x) + \phi'(y)$$

and

$$\frac{\partial q}{\partial x} = f'(x) + \phi'(y)$$

Hence,

$$\frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}$$

Since

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}$$

we get

$$\frac{\partial^2 z}{\partial y\, \partial x} = \frac{\partial^2 z}{\partial x\, \partial y}$$

To obtain the required PDE, differentiate z twice:

$$\frac{\partial^2 z}{\partial x \partial y} = f'(x) + \phi'(y)$$

Now from

$$p = yf'(x) + \phi(y), \qquad q = f(x) + x\phi'(y)$$

eliminate f, ϕ by forming

$$x\frac{\partial p}{\partial x} = y\frac{\partial q}{\partial y}$$

That is,

$$x\frac{\partial^2 z}{\partial x^2} = y\frac{\partial^2 z}{\partial y^2}$$

Therefore, the required PDE is

$$\boxed{x\,\frac{\partial^2 z}{\partial x^2} = y\,\frac{\partial^2 z}{\partial y^2}}$$

or equivalently,

$$\boxed{xz_{xx} - yz_{yy} = 0}$$

6. Form the PDE by eliminating arbitrary functions from the relation
$f(x^2+y^2+z^2, z^2-2xy)=0.$

Solution: Given

$$f(x^{2},+y^2+z^2\ z^2-2xy)=0$$

where f is an arbitrary function.

Let

$$u=x^2+y^2+z^2, v=z^2-2xy.$$

Since $f(u, v)=0,$ u and v are functionally dependent.

Therefore,

$$\frac{\partial(u, v)}{\partial(x, y)}=0.$$

Let

$$p=\frac{\partial z}{\partial x}, q=\frac{\partial z}{\partial y}.$$

Step 1: Compute the partial derivatives

For $u = x^2 + y^2 + z^2$,

$$u_x = 2x + 2zp, u_y = 2y + 2zq.$$

For $v = z^2 - 2xy$,

$$v_x = 2zp - 2y, v_y = 2zq - 2x.$$

Step 2: Form the Jacobian

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = 0.$$

Substituting the derivatives,

$$(2x + 2zp)(2zq - 2x) - (2y + 2zq)(2zp - 2y) = 0.$$

Dividing by 4,

$$(x + zp)(zq - x) - (y + zq)(zp - y) = 0.$$

On expanding,

$$xzq - x^2 + z^2pq - xzp - yzp + y^2 - z^2pq + yzq = 0.$$
$$(x+y)zq - (x+y)zp + y^2 - x^2 = 0.$$
$$(x+y)z(q-p) + (y-x)(x+y) = 0.$$
$$(x+y)[z(q-p) + (y-x)] = 0.$$

Hence the required first-order PDE is

$$\boxed{z(q-p) = x - y}$$

or, in derivative notation,

$$\boxed{z\left(\frac{\partial z}{\partial y} - \frac{\partial z}{\partial x}\right) = x - y.}$$

7. Form the PDE by eliminating arbitrary functions in the relation $f(x^2+y^2, x^2- z^2)=0$.

Solution: Given

$$f(x^2, +y^2\ x^2-z^2)=0$$

where f is an arbitrary function.

Let

$$u=x^2+y^2, v=x^2-z^2.$$

Since $f(u, v)=0$, $u$and $v$are functionally dependent. Hence,

$$\frac{\partial(u, v)}{\partial(x, y)}=0.$$

Let

$$p=\frac{\partial z}{\partial x}, q=\frac{\partial z}{\partial y}.$$

Step 1: Compute the partial derivatives

For $u = x^2 + y^2$,

$$u_x = 2x, u_y = 2y.$$

For $v = x^2 - z^2$,

$$v_x = 2x - 2zp, v_y = -2zq.$$

Step 2: Form the Jacobian

$$\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = 0.$$

Substituting,

$$\begin{vmatrix} 2x & 2y \\ 2x - 2zp & -2zq \end{vmatrix} = 0.$$

On expanding,

$$(2x)(-2zq) - 2y(2x - 2zp) = 0.$$
$$-4xzq - 4xy + 4yzp = 0.$$

Dividing by 4,

$$yzp - xzq - xy = 0.$$

Therefore, the required PDE is

$$\boxed{yz\,p - xz\,q = xy}$$

or, equivalently,

$$\boxed{yz\frac{\partial z}{\partial x} - xz\frac{\partial z}{\partial y} = xy}.$$

8. Solve and conclude the solution of PDE $\frac{\partial^2 u}{\partial x^2} = x + y$.

Solution: We are given the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} = x + y.$$

To find $u(x, y)$, integrate twice with respect to x, treating y as a constant.

Step 1: First integration

$$\frac{\partial u}{\partial x} = \int (x + y)\, dx = \frac{x^2}{2} + xy + \phi_1(y),$$

where $\phi_1(y)$ is an arbitrary function of y.

Step 2: Second integration

$$u = \int \left(\frac{x^2}{2} + xy + \phi_1(y)\right) dx = \frac{x^3}{6} + \frac{x^2 y}{2} + x\phi_1(y) + \phi_2(y),$$

where $\phi_2(y)$ is another arbitrary function of y.

Verification

$$\frac{\partial u}{\partial x} = \frac{x^2}{2} + xy + \phi_1(y),$$

and

$$\frac{\partial^2 u}{\partial x^2} = x + y,$$

which satisfies the given PDE.

Complete (General) Solution

$$\boxed{u(x, y) = \frac{x^3}{6} + \frac{x^2 y}{2} + x\,\phi_1(y) + \phi_2(y)}$$

where $\phi_1(y)$ and $\phi_2(y)$ are arbitrary functions of y.

Conclusion

The PDE

$$\frac{\partial^2 u}{\partial x^2} = x + y$$

has infinitely many solutions. Its general solution consists of a particular solution

$$\frac{x^3}{6} + \frac{x^2 y}{2},$$

Plus, the complementary part

$$x\,\phi_1(y) + \phi_2(y),$$

containing two arbitrary functions of y.

9. Solve and conclude the solution of PDE $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a$.

Solution: Given the PDE

$$\frac{\partial^2 z}{\partial x\,\partial y} = \frac{x}{y} + a,$$

where a is a constant.

Let

$$p = \frac{\partial z}{\partial x}.$$

Then

$$\frac{\partial p}{\partial y} = \frac{x}{y} + a.$$

Step 1: Integrate with respect to y

$$p = \int \left(\frac{x}{y} + a\right) dy = x \ln y + ay + \phi(x),$$

where $\phi(x)$ is an arbitrary function of x.

Since $p = \frac{\partial z}{\partial x}$,

$$\frac{\partial z}{\partial x} = x \ln y + ay + \phi(x).$$

Step 2: Integrate with respect to x

$$z = \int \big(x \ln y + ay + \phi(x)\big)\, dx.$$

Treating y as constant,

$$z = \frac{x^2}{2} \ln y + axy + \int \phi\,(x)\, dx + \psi(y),$$

where $\psi(y)$ is an arbitrary function of y.

Let

$$F(x) = \int \phi\,(x)\,dx.$$

Hence,

$$\boxed{z = \frac{x^2}{2}\ln y + axy + F(x) + \psi(y)}$$

where $F(x)$ and $\psi(y)$ are arbitrary functions.

Verification

$$\frac{\partial z}{\partial x} = x\ln y + ay + F'(x),$$

and therefore

$$\frac{\partial^2 z}{\partial y\,\partial x} = \frac{x}{y} + a,$$

which matches the given PDE.

Conclusion

The complete solution of

$$\frac{\partial^2 z}{\partial x\,\partial y} = \frac{x}{y} + a$$

is

$$\boxed{z = \frac{x^2}{2}\ln y + axy + F(x) + G(y)}$$

where $F(x)$ and $G(y)$ are arbitrary functions. The terms $F(x)$ and $G(y)$ represent the complementary part of the solution, while

$$\frac{x^2}{2}\ln y + axy$$

is a particular integral.

10. Find the solution of PDE $\frac{\partial^2 z}{\partial x \partial y} = \sin x$ sin y for which $\frac{\partial z}{\partial y} = -2\sin y$ when $x = 0$ and $z = 0$ if y is an odd multiple of $\frac{\pi}{2}$. $\left[or\ z = 0\ if\ y = (2n+1)\frac{\pi}{2}\right]$.

Solution: We are given

$$\frac{\partial^2 z}{\partial x\, \partial y} = \sin x \sin y\,,$$

with the conditions

$$\frac{\partial z}{\partial y} = -2\sin y\ \ when\ x = 0,$$

and

$$z = 0\ when\ y = \frac{(2n+1)\pi}{2}.$$

Step 1: Integrate with respect to x

$$\frac{\partial z}{\partial y} = \int \sin x \sin y\ dx = -\cos x\ \ \sin y + \phi(y),$$

where $\phi(y)$ is an arbitrary function of y.

Using the condition

$$\frac{\partial z}{\partial y} = -2\sin y \;\; at\; x = 0,$$

we get

$$-\cos 0 \;\; \sin y + \phi(y) = -\sin y + \phi(y) = -2\sin y\,.$$

Hence,

$$\phi(y) = -\sin y\,.$$

Therefore,

$$\frac{\partial z}{\partial y} = -(\cos x + 1)\sin y\,.$$

Step 2: Integrate with respect to y

$$z = \int -(\cos x + 1)\sin y \;\, dy = (\cos x + 1)\cos y + F(x),$$

where $F(x)$ is an arbitrary function of x.

Step 3: Apply the second condition

Given

$$z = 0 \; when \; y = \frac{(2n+1)\pi}{2}.$$

Since

$$\cos\left(\frac{(2n+1)\pi}{2}\right) = 0,$$

the solution becomes

$$0 = (\cos x + 1) \cdot 0 + F(x).$$

Thus,

$$F(x) = 0.$$

Final Solution

$$\boxed{z = (1 + \cos x)\cos y}$$

Verification

$$\frac{\partial z}{\partial y} = -(1 + \cos x) \sin y \,,$$

and

$$\frac{\partial^2 z}{\partial x \, \partial y} = \sin x \;\; \sin y \,,$$

which satisfies the PDE.

Also,

$$\left.\frac{\partial z}{\partial y}\right|_{x=0} = -(1 + 1) \sin y = -2 \sin y \,,$$

and

$$z = 0 \; when \; y = \frac{(2n + 1)\pi}{2} \,.$$

Hence all conditions are satisfied.

$$\boxed{z = (1 + \cos x) \cos y}$$

is the required solution.

11. Solve and conclude the solution of PDE $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y}$ subject to the conditions $\frac{\partial z}{\partial x} = log_e x$ when $y = 1 \; and \; z = 0 \; when \; x = 1.$

Solution: We are given the PDE

$$\frac{\partial^2 z}{\partial x \, \partial y} = \frac{x}{y},$$

subject to

$$\frac{\partial z}{\partial x} = \ln x \;\; when \; y = 1,$$

and

$$z = 0 \; when \; x = 1.$$

Step 1: Integrate the PDE with respect to y

$$\frac{\partial z}{\partial x} = \int \frac{x}{y} \, dy = x \ln y + \phi(x),$$

where $\phi(x)$ is an arbitrary function of x.

Step 2: Use the condition $\frac{\partial z}{\partial x} = \ln x$ when $y = 1$

Since $\ln 1 = 0$,

$$\left.\frac{\partial z}{\partial x}\right|_{y=1} = x \ln 1 + \phi(x) = \phi(x).$$

But

$$\left.\frac{\partial z}{\partial x}\right|_{y=1} = \ln x \, .$$

Therefore,

$$\phi(x) = \ln x \, .$$

Hence

$$\frac{\partial z}{\partial x} = x \ln y + \ln x \, .$$

Step 3: Integrate with respect to x

$$z = \int (x \ln y + \ln x)\ dx.$$

Since y is constant with respect to x,

$$z = \frac{x^2}{2} \ln y + \int \ln x\ dx + \psi(y),$$

where $\psi(y)$ is an arbitrary function of y.

Using

$$\int \ln x\ dx = x \ln x - x,$$

we obtain

$$z = \frac{x^2}{2} \ln y + x \ln x - x + \psi(y).$$

Step 4: Apply the condition $z = 0$ when $x = 1$

Substituting $x = 1$,

$$0 = \frac{1}{2}\ln y + (1)\ln 1 - 1 + \psi(y).$$

Since $\ln 1 = 0$,

$$0 = \frac{1}{2}\ln y - 1 + \psi(y).$$

Thus,

$$\psi(y) = 1 - \frac{1}{2}\ln y\,.$$

Final Solution

Substituting $\psi(y)$,

$$\boxed{z = \frac{x^2}{2}\ln y + x\ln x - x + 1 - \frac{1}{2}\ln y}$$

or equivalently,

$$\boxed{z = \frac{x^2 - 1}{2}\ln y + x\ln x - x + 1.}$$

Verification

$$\frac{\partial z}{\partial x} = x \ln y + \ln x\,,$$

So, at $y = 1$,

$$\frac{\partial z}{\partial x} = \ln x\,.$$

Also,

$$\frac{\partial^2 z}{\partial x\,\partial y} = \frac{\partial}{\partial y}(x \ln y + \ln x) = \frac{x}{y}.$$

And at $x = 1$,

$$z = \frac{1-1}{2}\ln y + 1 \cdot \ln 1 - 1 + 1 = 0.$$

Hence all conditions are satisfied.

$$\boxed{z = \frac{x^2-1}{2}\ln y + x \ln x - x + 1}$$

is the required solution.

12. Find the solution of PDE: $p\ tanx + q\ tany = tanz$.

Solution: Given the first-order Lagrange PDE

$$p \tan x + q \tan y = \tan z \,,$$

Where,

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}.$$

This is of the form

$$Pp + Qq = R,$$

with

$$P = \tan x \,, \qquad Q = \tan y \,, \qquad R = \tan z \,.$$

Step 1: Form the auxiliary equations

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}.$$

Since

$$\frac{1}{\tan \theta} = \cot \theta \,,$$

the auxiliary equations become

$$\cot x \ \ dx = \cot y \ \ dy = \cot z \ \ dz.$$

Step 2: Obtain the first integral

From

$$\frac{dx}{\tan x} = \frac{dy}{\tan y},$$

$$\int \cot x \ dx = \int \cot y \ dy.$$

Therefore,

$$\ln(\sin x) - \ln(\sin y) = c_1,$$

or

$$\boxed{\frac{\sin x}{\sin y} = c_1}.$$

Step 3: Obtain the second integral

From

$$\frac{dy}{\tan y} = \frac{dz}{\tan z},$$

$$\int \cot y \; dy = \int \cot z \; dz.$$

Hence,

$$\ln(\sin y) - \ln(\sin z) = c_2,$$

or

$$\boxed{\frac{\sin y}{\sin z} = c_2}.$$

Step 4: Write the complete solution

The complete integral of a Lagrange PDE is

$$F(c_1, c_2) = 0.$$

Thus,

$$F\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0.$$

Hence the required general solution is

$$\boxed{F\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0,}$$

where F is an arbitrary function.

Equivalently, it may be written as

$$\boxed{\frac{\sin x}{\sin y} = C_1, \qquad \frac{\sin y}{\sin z} = C_2.}$$

These two independent integrals give the complete solution of the PDE.

13. Solve $\frac{y^2 z}{x} \,. p + xzq = y^2$.

Solution: Given the Lagrange first-order PDE

$$\frac{y^2 z}{x}\, p + xz\, q = y^2,$$

Where,

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}.$$

Step 1: Form the auxiliary equations

Comparing with

$$Pp + Qq = R,$$

we have

$$P = \frac{y^2 z}{x}, \qquad Q = xz, \qquad R = y^2.$$

Therefore,

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2}.$$

Step 2: Find the first integral

Choose multipliers

$$\left(\frac{1}{x}, -\frac{y}{x^2}, 0\right)$$

Then

$$\frac{1}{x}P - \frac{y}{x^2}Q + 0 \cdot R = \frac{y^2 z}{x^2} - \frac{y^2 z}{x^2} = 0.$$

Hence

$$\frac{1}{x}\,dx - \frac{y}{x^2}\,dy = 0.$$

Multiplying by x^2,

$$x\,dx - y\,dy = 0.$$

Integrating,

$$\frac{x^2}{2} - \frac{y^2}{2} = C_1,$$

or

$$\boxed{x^2 - y^2 = C_1}.$$

Step 3: Find the second integral

From the auxiliary equations,

$$\frac{dy}{xz} = \frac{dz}{y^2}.$$

Thus,

$$y^2\,dy = xz\,dz.$$

Along a characteristic, $x^2 - y^2 = C_1 = a$, so

$$x^2 = y^2 + a.$$

Hence

$$\frac{dy}{\sqrt{y^2 + a}\,z} = \frac{dz}{y^2}.$$

Using

$$\frac{dy}{xz} = \frac{dz}{y^2} \implies y^2 dy = xz\,dz,$$

and substituting $x^2 = y^2 + a$,

$$y^2 dy = z\sqrt{y^2 + a}\,dz.$$

Integrating,

$$\int \frac{y^2}{\sqrt{y^2+a}}\,dy = \int z\ dz.$$

Using

$$\int \frac{y^2}{\sqrt{y^2+a}}\,dy = \frac{1}{2}y\sqrt{y^2+a} - \frac{a}{2}\ln\ \left|y+\sqrt{y^2+a}\right|,$$

we get

$$\frac{1}{2}y\sqrt{y^2+a} - \frac{a}{2}\ln\ \left|y+\sqrt{y^2+a}\right| = \frac{z^2}{2} + C_2.$$

Since $a = x^2 - y^2$and $\sqrt{y^2+a} = x$,

$$\boxed{xy - (x^2 - y^2)\ln|x+y| - z^2 = C_2.}$$

Step 4: Complete solution

The complete solution of the Lagrange PDE is

$$F(C_1, C_2) = 0,$$

therefore

$$\boxed{F\left(x^2, -y^2\ \ xy - (x^2 - y^2)\ln|x + y| - z^2\right) = 0,}$$

where F is an arbitrary function.

This is the required complete integral of the PDE

$$\frac{y^2 z}{x}\, p + xz\, q = y^2.$$

14. Solve $(mz - ny)p + (nx - lz)q = (ly - mx)$.

Solution: We need to solve the first-order Lagrange PDE

$$(mz - ny)p + (nx - lz)q = (ly - mx),$$

Where,

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}.$$

Step 1: Form the auxiliary equations

The given PDE is of the form

$$Pp + Qq = R,$$

with

$$P = mz - ny, \qquad Q = nx - lz, \qquad R = ly - mx.$$

Hence the auxiliary equations are

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}.$$

Step 2: Find the first integral

Choose multipliers l, m, n.

Then

$$lP + mQ + nR = l(mz - ny) + m(nx - lz) + n(ly - mx) = 0.$$

Therefore,

$$l\,dx + m\,dy + n\,dz = 0.$$

Integrating,

$$lx + my + nz = C_1.$$

This gives the first integral:

$$u = lx + my + nz.$$

Step 3: Find the second integral

Choose multipliers x, y, z.

Then

$$xP + yQ + zR = x(mz - ny) + y(nx - lz) + z(ly - mx) = 0.$$

Hence

$$x\,dx + y\,dy + z\,dz = 0.$$

Integrating,

$$\frac{x^2 + y^2 + z^2}{2} = C_2,$$

or

$$x^2 + y^2 + z^2 = C_2.$$

Thus the second integral is

$$v = x^2 + y^2 + z^2.$$

Step 4: Write the complete solution

The complete integral of a Lagrange PDE is

$$F(u, v) = 0,$$

where u and v are two independent first integrals.

Therefore,

$$\boxed{F\,(\mathrm{lx} + \mathrm{my} + \mathrm{nz}, x^2 + y^2 + z^2) = 0}$$

is the complete solution.

Equivalently,

$$\boxed{lx + my + nz = \Phi\,(x^2 + y^2 + z^2)}$$

where Φ is an arbitrary function.

15. $Solve\ x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2).$

Solution: Solve the Lagrange first-order PDE

$$x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2),$$

where

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}.$$

Step 1: Form the auxiliary equations

$$\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)}.$$

Let

$$P = x(y^2 - z^2), \qquad Q = y(z^2 - x^2), \qquad R = z(x^2 - y^2).$$

Step 2: Obtain the first integral

Choose multipliers

$$\frac{1}{x}, \frac{1}{y}, \frac{1}{z}.$$

Then

$$\frac{P}{x} + \frac{Q}{y} + \frac{R}{z} = (y^2 - z^2) + (z^2 - x^2) + (x^2 - y^2) = 0.$$

Hence

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0.$$

Integrating,

$$\ln x + \ln y + \ln z = C_1,$$

or

$$\boxed{xyz = C_1}.$$

Step 3: Obtain the second integral

Choose multipliers

$$x, y, z.$$

Then

$$xP + yQ + zR = x^2(y^2 - z^2) + y^2(z^2 - x^2) + z^2(x^2 - y^2) = 0.$$

Therefore,

$$x\,dx + y\,dy + z\,dz = 0.$$

Integrating,

$$\frac{x^2 + y^2 + z^2}{2} = C_2,$$

or

$$\boxed{x^2 + y^2 + z^2 = C_2}.$$

Step 4: Complete solution

The complete integral of the Lagrange PDE is

$$F(C_1, C_2) = 0.$$

Therefore,

$$\boxed{F\,(x, yz\ \ x^2 + y^2 + z^2) = 0}$$

or equivalently,

$$\boxed{xyz = \Phi\,(x^2 + y^2 + z^2)},$$

where Φ is an arbitrary function.

Hence the required solution is

$$\boxed{F\,(x, yz\ \ x^2 + y^2 + z^2) = 0.}$$

16. Solve by variable separable method $4\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}=3u$,given that $u(0,y)=2e^{5y}$.

Solution: We need to solve

$$4\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}=3u,$$

subject to

$$u(0,y)=2e^{5y}.$$

Using the **method of separation of variables**, assume

$$u(x,y)=X(x)Y(y).$$

Then

$$u_x=X'Y, \qquad u_y=XY'.$$

Substituting into the PDE,

$$4X'Y+XY'=3XY.$$

Dividing by XY,

$$4\frac{X'}{X}+\frac{Y'}{Y}=3.$$

Rearrange:

$$4\frac{X'}{X} - 3 = -\frac{Y'}{Y} = k,$$

where k is a separation constant.
Thus, we obtain two ODEs:

$$4\frac{X'}{X} = 3 + k,$$

and

$$\frac{Y'}{Y} = -k.$$

Solve for $X(x)$

$$\frac{X'}{X} = \frac{3 + k}{4}.$$

Integrating,

$$X = A\, e^{\frac{3+k}{4}x}.$$

Solve for $Y(y)$

$$\frac{Y'}{Y} = -k.$$

Integrating,

$$Y = B\, e^{-ky}.$$

Form the separated solution

$$u = C\, e^{\frac{3+k}{4}x - ky},$$

where $C = AB$.

Using the boundary condition $u(0, y) = 2e^{5y}$,

$$Ce^{-ky} = 2e^{5y}.$$

Comparing exponents,

$$-k = 5 \;\Rightarrow\; k = -5,$$

and

$$C = 2.$$

Substituting $k = -5$,

$$u(x, y) = 2e^{\frac{3-5}{4}x+5y} = 2e^{-\frac{x}{2}+5y}.$$

Verification

$$u_x = -e^{-\frac{x}{2}+5y}, \qquad u_y = 10e^{-\frac{x}{2}+5y}.$$

Hence

$$4u_x + u_y = (-4 + 10)e^{-\frac{x}{2}+5y} = 6e^{-\frac{x}{2}+5y} = 3u.$$

Also,

$$u(0, y) = 2e^{5y}.$$

Therefore, the solution is

$$\boxed{u(x, y) = 2e^{5y-\frac{x}{2}}}.$$

17. Solve $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$ by the method of variable separable.

Solution: Consider the PDE

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0.$$

We solve it by the **method of separation of variables**.

Step 1: Assume a separable solution

Let

$$u(x, y) = X(x)Y(y).$$

Then

$$u_x = X'Y, \qquad u_y = XY'.$$

Substituting into the PDE,

$$x^2 X'Y + y^2 XY' = 0.$$

Dividing by XY,

$$x^2 \frac{X'}{X} + y^2 \frac{Y'}{Y} = 0.$$

Rearrange:

$$x^2\frac{X'}{X} = -y^2\frac{Y'}{Y} = k,$$

where k is a separation constant.

Thus,

$$x^2\frac{X'}{X} = k, y^2\frac{Y'}{Y} = -k.$$

Step 2: Solve for $X(x)$

$$\frac{X'}{X} = \frac{k}{x^2}.$$

Integrating,

$$\ln X = k\int x^{-2}\,dx = -\frac{k}{x} + C_1.$$

Hence

$$X = Ae^{-k/x}.$$

Step 3: Solve for $Y(y)$

$$\frac{Y'}{Y} = -\frac{k}{y^2}.$$

Integrating,

$$\ln Y = -k \int y^{-2}\, dy = \frac{k}{y} + C_2.$$

Hence

$$Y = Be^{k/y}.$$

Step 4: Form the solution

$$u = XY = AB\, e^{-k/x + k/y}.$$

Let $C = AB$. Then

$$u = C\, e^{k\left(\frac{1}{y} - \frac{1}{x}\right)}.$$

Since k is arbitrary, the complete solution can be written as an arbitrary function of the combination

$$\frac{1}{x} - \frac{1}{y}.$$

Therefore,

$$\boxed{u = \Phi\left(\frac{1}{x} - \frac{1}{y}\right)},$$

where Φ is an arbitrary function.

Answer

$$\boxed{u(x, y) = \Phi\left(\frac{1}{x} - \frac{1}{y}\right)}.$$

This is the general solution obtained by the method of separation of variables.

18. Solve by the method of variable separable $\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = u$,where $u = 6e^{-3x}$ when $y = 0$.

Solution: We are given the PDE

$$\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = u,$$

with the condition

$$u(x, 0) = 6e^{-3x}.$$

We solve it by the **method of separation of variables**.

Step 1: Assume a separable solution

Let

$$u(x, y) = X(x)Y(y).$$

Then

$$u_x = X'Y, \qquad u_y = XY'.$$

Substituting into the PDE,

$$X'Y + 2XY' = XY.$$

Dividing by XY,

$$\frac{X'}{X} + 2\frac{Y'}{Y} = 1.$$

Separate the variables:

$$\frac{X'}{X} - 1 = -2\frac{Y'}{Y} = k,$$

where k is a separation constant.

Thus,

$$\frac{X'}{X} = 1 + k,$$

and

$$\frac{Y'}{Y} = -\frac{k}{2}.$$

Step 2: Solve for $X(x)$

$$\frac{X'}{X} = 1 + k.$$

Integrating,

$$X = Ae^{(1+k)x}.$$

Step 3: Solve for $Y(y)$

$$\frac{Y'}{Y} = -\frac{k}{2}.$$

Integrating,

$$Y = Be^{-ky/2}.$$

Step 4: Form the separated solution

$$u = C\, e^{(1+k)x - \frac{k}{2}y},$$

where $C = AB$.

Step 5: Apply the condition $u(x, 0) = 6e^{-3x}$

At $y = 0$,

$$u(x, 0) = Ce^{(1+k)x} = 6e^{-3x}.$$

Comparing coefficients and exponents,

$$C = 6,$$

and

$$1 + k = -3 \;\Rightarrow\; k = -4.$$

Step 6: Obtain the required solution

Substituting $k = -4$,

$$u(x, y) = 6e^{(-3)x - \frac{-4}{2}y} = 6e^{-3x+2y}.$$

Verification

$$u_x = -18e^{-3x+2y}, u_y = 12e^{-3x+2y}.$$

Therefore,

$$u_x + 2u_y = (-18 + 24)e^{-3x+2y} = 6e^{-3x+2y} = u.$$

Also,

$$u(x, 0) = 6e^{-3x}.$$

Hence the solution is

$$\boxed{u(x, y) = 6e^{-3x+2y}}.$$

19. Obtain various possible solution and hence find the suitable solution of one-dimensional wave equation by variable separable method.

Solution: Consider the **one-dimensional wave equation**

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2},$$

where c is the wave velocity.

Solution by the Method of Separation of Variables

Assume

$$u(x, t) = X(x)T(t).$$

Then

$$u_{tt} = XT'', u_{xx} = X''T.$$

Substituting into the wave equation,

$$XT'' = c^2 X''T.$$

Dividing by XT,

$$\frac{T''}{T} = c^2 \frac{X''}{X}.$$

Since the left side depends only on t and the right side only on x, both must be equal to a constant, say k:

$$\frac{T''}{T} = c^2\frac{X''}{X} = k.$$

Thus

$$X'' - \frac{k}{c^2}X = 0, T'' - kT = 0.$$

The nature of the solution depends on the sign of k.

Case I: $k = p^2 > 0$

Then

$$X'' - \frac{p^2}{c^2}X = 0,$$

$$T'' - p^2T = 0.$$

Hence

$$X = Ae^{px/c} + Be^{-px/c},$$

$$T = Ce^{pt} + De^{-pt}.$$

Therefore

$$\boxed{u = \left(Ae^{px/c} + Be^{-px/c}\right)(Ce^{pt} + De^{-pt})}.$$

Case II: $k = 0$

Then

$$X'' = 0, T'' = 0.$$

Integrating,

$$X = A + Bx,$$
$$T = C + Dt.$$

Hence

$$\boxed{u = (A + Bx)(C + Dt)}.$$

Case III: $k = -p^2 < 0$

Then

$$X'' + \frac{p^2}{c^2} X = 0,$$
$$T'' + p^2 T = 0.$$

The solutions are

$$X = A \cos\frac{px}{c} + B \sin\frac{px}{c},$$
$$T = C \cos p\,t + D \sin p\,t.$$

Thus

$$\boxed{u = \left(A \cos\frac{px}{c} + B \sin\frac{px}{c}\right)(C \cos p\,t + D \sin p\,t)}.$$

Suitable Solution

For most physical vibration problems (string fixed at boundaries), the solution must remain bounded for all $x$and t.

- Case I contains exponential terms and generally becomes unbounded.
- Case II gives only trivial linear solutions.
- Case III gives bounded oscillatory solutions representing waves.

Hence the **physically suitable separated solution** is obtained from **Case III**:

$$\boxed{u(x, t) = \left(A\cos\frac{px}{c} + B\sin\frac{px}{c}\right)(C\cos p\,t + D\sin p\,t)}.$$

or equivalently,

$$\boxed{u(x, t) = \left(A\cos\frac{px}{c} + B\sin\frac{px}{c}\right)(C_1\cos p\,t + C_2\sin p\,t).}$$

This is the required suitable solution of the one-dimensional wave equation obtained by the method of separation of variables.

20. Solve the wave equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ by the method of variable separable.

Solution: Consider the PDE

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

which is the **two-dimensional Laplace equation**.

Step 1: Assume a separable solution

Let

$$u(x, y) = X(x)Y(y).$$

Then

$$u_{xx} = X''(x)Y(y), \quad u_{yy} = X(x)Y''(y).$$

Substituting into the PDE,

$$X''Y + XY'' = 0.$$

Dividing by XY,

$$\frac{X''}{X} + \frac{Y''}{Y} = 0.$$

Hence

$$\frac{X''}{X} = -\frac{Y''}{Y} = k,$$

where $k$is a separation constant.

Thus, we obtain

$$X'' - kX = 0, \qquad Y''+kY = 0.$$

Step 2: Solve for different values of k

Case I: $k = p^2 > 0$

Then

$$X'' - p^2 X = 0, Y'' + p^2 Y = 0.$$

Their solutions are

$$X = Ae^{px} + Be^{-px},$$
$$Y = C\cos(py) + D\sin(py).$$

Therefore

$$\boxed{u = (Ae^{px} + Be^{-px})(C\cos p\,y + D\sin p\,y)}.$$

Case II: $k = 0$

Then

$$X'' = 0, Y'' = 0.$$

Hence

$$\begin{gathered} X = A + Bx, \\ Y = C + Dy. \end{gathered}$$

Thus

$$\boxed{u = (A + Bx)(C + Dy).}$$

Case III: $k = -p^2 < 0$

Then

$$X'' + p^2 X = 0, Y'' - p^2 Y = 0.$$

Hence

$$X = A\cos(px) + B\sin(px),$$

$$Y = Ce^{py} + De^{-py}.$$

Therefore

$$\boxed{u = (A\cos p\,x + B\sin p\,x)(Ce^{py} + De^{-py})}.$$

General Separable Solutions

The Laplace equation admits the following separated forms:

$$\boxed{u = (Ae^{px} + Be^{-px})(C\cos p\,y + D\sin p\,y)}$$

or

$$\boxed{u = (A\cos p\,x + B\sin p\,x)(Ce^{py} + De^{-py})}$$

together with the special case

$$\boxed{u = (A + Bx)(C + Dy).}$$

These are the solutions obtained by the **method of separation of variables**.